CS 203 Assignment 1, Project 4: a course advisory dataset for IIT Gandhinagar M.Tech students

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1 Overview

We built a dataset for a small language model that answers M.Tech course advising questions: what is still left for my degree, can I register for a course next term, and which electives should I take. Scope is four programmes in the CS and EE departments, MTech-CSE, MTech-AI, MTech-EE and MTech-ICDT, each with a thesis and a coursework track (`docs/DECISIONS.md` D1 to D3). A rule engine built from the institute's own documents computes every gold answer, so each label points back to a source page. A local LLM (Qwen2.5-1.5B-Instruct) paraphrases questions and justifications and two humans audit those paraphrases. Headline numbers: 104 courses and 77 rules with provenance, 180 synthetic students, 600 advisory pairs (200 per intent), 1526 verified augmented children, 80 doubly labelled audit items, a frozen gold test set of 76 items, and 10 real Google Form questions as a shifted test set.

2 Part A: Collection

Sources and provenance. The raw layer holds eight academic advisories (Adv5, Adv8, Adv17, Adv24, Adv42, Adv60, Adv_05, Adv_08), the revised CSE M.Tech course structure 2025-26, the Sem-I 2026-27 timetable (xlsx) and four pages of the EE M.Tech Google Site. Each file has a sha256 in `data/raw/sources/source_metadata.json`. Every course and rule record carries `source_doc` plus `source_page`, `source_row` or `source_section`, and a `record_sha256`; every rule also stores a verbatim quote that `build_rules` checks against the page text. `data/metadata.csv` has 3074 rows (578 raw, 2496 processed), one per sample, with source, page, collector, tool, version and content hash: source files, PDF pages, timetable rows, course plans, raw and clean catalog records, every rule with a stated value (76 of 77), 180 students, 600 pairs, 1526 augmented children and 10 form responses.

Course plans. Plan links come from the timetable's "Link To Course Plan" cell. For the 50 CS/ES codes, 15 plans downloaded, 12 links need an IITGN login (Google sign-in page, not bypassed) and 23 codes have no link. For the 25 EE codes, 7 downloaded, 4 are login-gated and 14 have no link. That gives 22 plan files in `data/raw/course_plans/`.

Catalog and rules. `catalog_raw.json` keeps 136 records with their defects (duplicates, the "Algrithms" typo, float credits, section suffixes). `clean_catalog` turns them into 104 courses (67 basket, 15 open elective, 6 stream core, 2 compulsory, 6 requirement, 8 UG background) and logs all 78 changes in `cleaning_changes.csv`. `rules.json` has 77 rules, 20 of them EE-specific (`ee.*`); one, maximum credits per semester, is marked `not_stated` because no source gives it.

Prerequisites finding. No source states an enforceable PG prerequisite: 103 of 104 courses have `prereq_source: none_stated` and one has a manual entry with an empty list (CS 613, "Prerequisite (Optional)" with UG courses "or equivalent"; first proposed by the pipeline assistant, confirmed by hem436). So the engine has two tiers: hard prerequisites (empty in practice, but checked) and `recommended_background`, which 3 courses carry. The prerequisite graph has 104 nodes and 9 edges, all of type recommended, no missing codes and no cycles. D4 stores UG codes named in plans as recommended background; D5 (decided by hem436) says UG-level background never warns an M.Tech student, because they did their UG elsewhere.

Synthetic students, consent, data flow. The engine generates 180 students from seed 203 (CSE 64, EE 51, AI 41, ICDT 24; 92 thesis, 88 coursework) with 1422 transcript entries and 0 rule violations. The institute documents are public regulations used for non-commercial coursework and are not redistributed; course plans were saved only where the link opened without a login, and instructor names are not copied. Form respondents tick a required consent box, no emails or names are collected, the Google timestamp

is dropped and contact details in free text are removed (`docs/CONSENT_AND_LICENSING.md`). Data flows sources → `extract_sources` → `fetch_course_plans/fetch_ee_pages` → `build_catalog` → `clean_catalog` and `build_rules` → rule engine → queries → Label Studio (`docs/DATA_FLOW.md`).

3 Part B: Validation

The suite has 25 checks (V01 to V25) written with Pandera schemas, networkx graph checks, rapidfuzz near-duplicate titles and the rule engine itself. It runs on three stages. On the raw catalog 10 checks apply: 7 fail, 2 pass, 1 warns. On processed data all 25 run: 24 pass and V10 warns about 2 near-duplicate titles of different codes, kept on purpose (Table 1). On a corrupted copy made by the DVC stage `validate_corrupted` (seed 209, fault list in its output `data/interim/corrupted/faults.json`) all 15 injected faults are caught, including a prerequisite cycle, an unknown prerequisite, total credits changed from 82 to 80, a below-600 cap breach and train/test leakage; one planned fault (per-semester credit cap) was not injected because no cap is stated.

Table 1: Before and after cleaning (failures; `supporting_files/B_validation/before_after.md`).

check	what it tests	raw	after
V02	code format	6	0
V03	credits are integers in range	49	0
V05	required fields present	9	0
V06 / V07	duplicate codes / exact duplicate records	5 / 4	0 / 0
V08 / V09	title matches timetable / title whitespace and punctuation	5 / 4	0 / 0
V10	near-duplicate titles (warn)	2	2

Validation alone did not catch everything; human review of the gold answers found two engine bugs. (1) While labelling the first item, hem436 saw "Other baskets (): 0 credits earned" for an MTech-AI student: the engine took other baskets from the student's own programme, which for MTech-AI has one stream. The fix follows the CSE structure (MTech-AI follows the AI-ML curriculum) and changed 100 of 600 gold answers, all MTech-AI. (2) hem436 noticed eligibility answers telling M.Tech students they lacked IITGN UG courses (for example ES 666 needing ES 112). D5 removed these warnings and changed 53 gold answers, 0 query texts. Both fixes have regression tests and happened before any finished annotation was affected. Because the AI fix also changes how course roles are assigned (it would have changed 12 of the 180 synthetic students, all MTech-AI), the transcript generator keeps the old basket view (`role_of(..., own_programme_baskets=True)`) so the labelled students are reproduced exactly, while all gold answers use the fixed rule.

4 Part C: Labelling and agreement

Task. Each item shows a student summary, the template query, the LLM paraphrase, the rule-engine gold answer and the LLM-rewritten justification. Annotators fill three levels in a fixed order: decision (accept, edit, reject), then one error category (none, hallucinated_course, wrong_fact_or_policy, meaning_drift, incomplete_justification, fluency, ambiguous_query), then the edit text. The decision comes first so the category list cannot steer it. Two annotators (A1 = SMH, A2 = MSB) labelled the same 40 items per round in Label Studio 1.23.2, two disjoint rounds, all drawn from the test split (seed 208). Round 1 used guidelines v1; after it we wrote v2 (seven rules, Appendix C), each tied to round-1 disagreements, and used it for round 2.

Table 2: Inter-annotator agreement, 40 items per round, bootstrap 95% CI with 2000 resamples.

	round 1 (guidelines v1)		round 2 (guidelines v2)	
level	Cohen's κ [95% CI]	raw agr.	Cohen's κ [95% CI]	raw agr.
decision	0.227 [0.000, 0.469]	65.0%	0.377 [0.094, 0.630]	77.5%
error category	0.171 [0.007, 0.353]	62.5%	0.350 [0.064, 0.620]	75.0%
disagreements	17		12	

Disagreements. Round 1 was mostly about strictness, not category confusion: A2 accepted 34 of

40 items and A1 24, and in 11 of the 17 disagreements A1 flagged an item A2 accepted (Figure 1). Causes: vague "almost there" justifications, a contradicted number inside fluent text, the edit-or-reject threshold, and A1 correcting a correct gold answer from memory. In round 2 the main split left was wrong numbers in fluent text, plus the paraphraser turning "coursework track" into "workshop track". Kappa rose in both levels, but the CIs overlap and the rounds use different items, so the gain is in the expected direction, not conclusive.

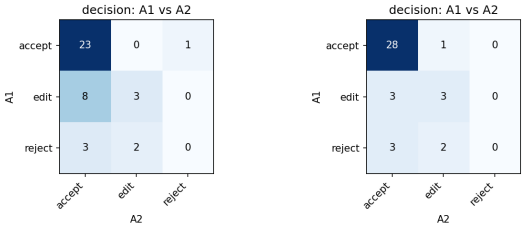


Figure 1: Decision confusion, A1 rows, A2 columns: round 1 (left), round 2 (right).

Adjudication. 50 items agreed on both levels and needed no adjudication. For the other 30 (17 + 13; round 2 also includes one item where both chose edit with different texts), the assistant drafted a proposed label and a gold-grounded edit for each of the 30 (the 29 disagreements plus that item), with a one-line reason each. A1 and A2 reviewed every proposal together on a review page and made the call on the 29 disagreements: 28 approved, 1 overridden (`decided_by` "A1+A2"). A1 decided the extra item. No item stayed unresolved. Final audit outcome over 80 items: 55 accept, 21 edit, 4 reject. The gold test set is the 76 accepted or edited items (27 eligibility, 27 elective recommendation, 22 remaining requirements), sha256 `cd2e08dcb56f5716b06981e096c3c93fd417abab1989b5d22d45befeafbc383f` in `data/gold/GOLD_SHA256.txt`.

Real form queries. 10 consenting responses were imported (0 excluded), 8 from form v1 and 2 from v2. v2 added a description and three example questions after respondents were unsure what to ask, so later replies may echo those examples; each row carries its `form_version`. Real questions differ from templates: they are informal ("sem 3 of m.tech EE,SPML stream"), give course counts instead of codes, and ask things outside the three intents (proficiency transcript, graduating by a date, other-department courses). By kind: 6 remaining requirements, 2 elective advice, 1 eligibility, 1 other.

5 Part D: Augmentation and active learning

Every augmented child is kept only if `src/augment/verify.py` confirms the rule engine gives the same answer (or, for transcript length, the correctly recomputed answer). LLM paraphrase children must also keep every fact the question states (programme, stream, track, semesters completed, number of electives, term) and must not turn the question back on the student (guidelines v2 R4). Of 1610 children, 1526 were kept.

Table 3: Augmentation invariance (`supporting_files/D_augmentation/invariance.csv`).

augmentation	check	children	pass	rate
LLM paraphrase	gold and slots unchanged	120	36	0.3
typos and abbreviations	gold unchanged	600	600	1.0
transcript order shuffled	gold unchanged	445	445	1.0
transcript length (earlier snapshot)	gold recomputed correctly	445	445	1.0

Only 36 of 120 LLM paraphrases pass: most drop or change a fact the question states. The most frequent lost slot is the track (in 47 failures, 26 of them track alone), then semesters completed (35) and stream (21); 10 turn the question to the student. This matches the "workshop track" errors the annotators caught in Part C. The typo and transcript checks pass at 1.0 by construction, since those transforms never touch the facts the engine reads (or, for length, the engine recomputes the answer). For active learning, the base model (same Qwen2.5-1.5B, greedy) answered the 336 train pairs; 232 of its answers fail against gold (wrong remaining 94, ineligible recommendation 64, wrong eligibility 56, invalid JSON 18, hallucinated code 2). The score ranks pairs by disagreement with the rule engine.

Table 4 gives the share of the 232 failures each strategy captures; random is the mean of 5 seeds. This is a label-efficiency comparison, not a learning curve: no model is retrained on the picks.

Table 4: Active learning vs random (`supporting_files/D_active_learning/label_efficiency.csv`).

strategy	budget 25	budget 50	budget 100	failure types	intents
AL top-k (disagreement)	0.11	0.22	0.43	0.60	1
AL stratified by intent (later)	0.10	0.19	0.38	1.00	3
random (5 seeds)	0.08	0.15	0.30	0.88 to 0.92	3

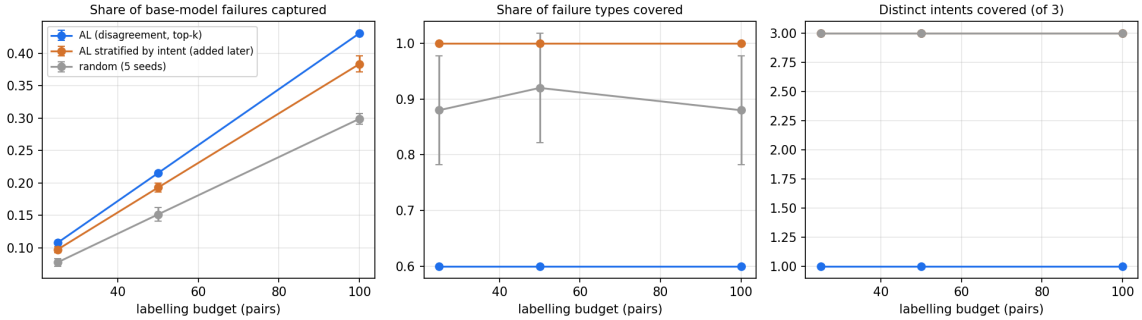


Figure 2: Label efficiency: failures captured, failure types covered and intents covered per budget.

Honest reading: the score and the failure test are the same comparison with the rule engine, so every top-k pick is a failure (precision 1 by construction) and its "failures captured" is just budget over 232. The informative comparison is which failures each strategy reaches. Plain top-k takes every pick from one intent (remaining requirements) and it covers only 0.60 of the failure types. We saw this after the first run and added an intent-stratified variant; it keeps most of the gain over random and covers all failure types and intents. The original strategy and its numbers are kept unchanged and the stratified one is reported as a later addition.

6 Part E: Versioning and splits

DVC tracks the frozen human and raw inputs with `dvc add` (`data/raw/sources`, both course-plan folders, the raw form CSV, the LLM cache `data/interim/llm_cache.jsonl`, and the round 1 and 2 tasks, manifests, exports and adjudication files) and the pipeline in `dvc.yaml` (26 stages, from `extract_sources` through the human-input stages to `handoff` and `validate_corrupted`; EE fetching is frozen). Evidence files are outputs with `cache: false` so they stay readable in git. `dvc.lock` pins the hash of every stage output. A full `dvc repro` runs offline (LLM stages replay the cache) and reproduced courses, rules, students, queries, paraphrases, augmented pairs, splits, audited labels, the gold set and base-model answers byte for byte; `data/metadata.csv` differs only in its `collected_at_utc` run time (`supporting_files/F_reproducibility/repro_check.txt`). The remote is a local folder `../dvc_remote`. The freeze tag `a1-final` named in `HANDOFF_A2.md` is created after the final form import (not yet set when this was built).

Students are assigned to splits first, per programme, by student count (seed 207), and every pair is then rendered with a template allowed in its split. One template per family (7 of 22) is test-only. All MTech-ICDT students form a programme holdout: it is the smallest programme and the only one without streams, under EE rules. Augmented children inherit their root parent's split. All overlap checks are zero (students across splits, train/val templates in test, ICDT in train/val/test, duplicate ids), and no pair was dropped.

The shifted split `data/splits/shifted_forms.csv` holds the real form questions with `template_family real_form`; it has no gold answers and is for a distribution-shift check only.

7 Part F: Report and reproducibility

`DATASET_CARD.md` lists sources, sizes, split balance, known biases and intended use (advisory model

Table 5: Split sizes: students assigned, original pairs, augmented children, total rows (`supporting_files/E_splits/split_summary.json`).

split	students	pairs	children	rows	eligible / not	pairs CSE / AI / EE / ICDT
train	101	336	869	1205	56 / 56	127 / 90 / 119 / 0
val	24	80	200	280	13 / 13	35 / 21 / 24 / 0
test	31	102	251	353	17 / 17	49 / 21 / 32 / 0
programme_holdout	24	82	206	288	14 / 14	0 / 0 / 0 / 82
shifted (forms)	n/a	10	0	10	n/a	see Part C

training and label-noise study, not official advice). `ENVIRONMENT.md` records the machine (Ubuntu 22.04.5, Ryzen 5 2500U, CPU only), Python 3.10.12, DVC 3.67.1, Label Studio 1.23.2 in its own venv, the model tag, digest and GGUF blob, all eight seeds, the three LLM backends and the Kaggle GPU (Tesla T4, CUDA 12.8, torch 2.11.0+cu128, transformers 5.16.1). `prompts/pipeline/README.md` records the model, versions and backends per prompt and the Qwen default system prompt. `LABELLING_PROTOCOL.md` records who labelled what, config and guideline per round, item order and seed, what annotators could see, the adjudication process and a timeline matched to export files. Exports are never edited. Rebuild: `python3.10 -m venv .venv && .venv/bin/pip install -r requirements.txt`, then `.venv/bin/dvc pull`, `.venv/bin/dvc repro` and `.venv/bin/pytest -q` (252 tests pass, `supporting_files/F_reproducibility/pytest.txt`). LLM answers are replayed from `data/interim/llm_cache.jsonl`, keyed by prompt id, input hash and model digest, so a rerun gives the same text.

8 Limitations

- Students are synthetic and valid by construction: no failed courses, substitutions or committee exceptions.
- The sources state no enforceable PG prerequisite, so eligibility mostly tests offering and already-taken rules.
- Kappa rests on 40 items per round; the CIs are wide and overlap.
- LLM backends are mixed: Ollama 0.1.38 could not load the qwen2 tokenizer, so the same GGUF ran on llama.cpp, and part of the paraphrases ran on a Kaggle T4 with transformers fp16. Each row records its backend (`para_backend`); outputs are not bit-identical across backends.
- The form examples were added mid-collection (v1 to v2), which may pull later wording toward them. We have 10 real questions, below the target of 30 at the time of writing.
- The EE site marks the ICDT discipline list "to be updated soon", so ICDT answers rest on a partial list.

Contributions

member	commits	work
Myank Singh Beniwal (MSB)	3	First data-loading commit ("Data loading and cleaning"); annotator A2 in rounds 1 and 2 (both exports committed from his own account); joint adjudication with A1 (all 29 reviewed decisions are `decided_by` A1+A2). The docs record form distribution by hem436 only.
Hemant (SMH, hem436)	34	Everything else: scope decisions, pipeline and rule engine (built with Claude Code), sources, form creation and sharing, annotator A1 in both rounds, joint adjudication, DVC, documentation and this report.

A Evidence index

Every number in the body, the file that backs it (paths from the repo root) and the field or row.

number	claim	file	field / row
4 programmes, 2 tracks	scope	`docs/DECISIONS.md`	D1, D2, D3
104	courses	`data/processed/courses.json`	list length; also `supporting_files/B_engine/prereq_graph_report.json` `nodes`
67 / 15 / 6 / 2 / 6 / 8	courses by category	`data/processed/courses.json`	`category` counts
77; 20; 1	rules; `ee.*` rules; not stated	`data/processed/rules.json`	`rules` keys; prefix `ee.`; `not_stated: true`
180	synthetic students	`supporting_files/B_engine/transcripts_summary.json`	`n_students`
64 / 51 / 41 / 24	students CSE / EE / AI / ICDT	same	`programmes`
92 / 88	thesis / coursework	same	`tracks`
1422; 0	transcript entries; violations	same	`transcript_entries`, `violations`
203	transcript seed	`params.yaml`	`seeds.transcripts`
600; 200 per intent	advisory pairs	`supporting_files/B_engine/queries_summary.json`	`n_pairs`, `per_intent`
1526; 1610	children kept; children in	`supporting_files/E_splits/split_summary.json`, `supporting_files/D_augmentation/verify_report.json`	`n_children_in` (sum of `child_rows_per_split`); `children_in`
80	audit items	`data/labels/audited.jsonl`	row count (40 with round 1, 40 with round 2)
76	gold items	`supporting_files/E_splits/gold_summary.json`	`n_gold`
10	real form questions	`supporting_files/C_labelling/forms_import_log.json`	`rows_kept`; `supporting_files/E_splits/shifted_summary.json` `n_rows`
0	excluded responses	`supporting_files/C_labelling/forms_import_log.json`	`rows_excluded`
8 / 2	form v1 / v2 replies	`data/raw/forms/responses_anonymised.jsonl`	`form_version` counts
6 / 2 / 1 / 1	replies by kind	`supporting_files/E_splits/shifted_summary.json`	`by_question_kind`
8 advisories, CSE structure, timetable, 4 EE pages	raw sources	`data/raw/sources/source_metadata.json`, `data/metadata.csv`	`sources[].sha256`; `artifact` rows under `data/raw/sources/`
3074 (578 raw, 2496 processed)	metadata rows	`data/metadata.csv`	row count; `layer` counts
180 / 600 / 1526 / 10 / 76	metadata rows for students, pairs, children, forms, rules	`data/metadata.csv`	`sample_id` prefixes `student`, `pair`, `aug`, `form`, `rule`
50: 15 / 12 / 23	CS/ES plan codes: downloaded / login / no link	`supporting_files/A_collection/course_plan_fetch_log.csv`	last `outcome` per `code` (ok, failed with "Google sign-in page", no_url)
25: 7 / 4 / 14	EE plan codes	`supporting_files/A_collection/ee_course_plan_fetch_log.csv`	same
22	plan files	`data/raw/course_plans/`	file count (15 + 7)

number	claim	file	field / row
136	raw catalog records	`data/metadata.csv`	rows with `artifact` = `data/raw/catalog_raw.json`; `supporting_files/A_collection/collection_log.md` section 7
78	cleaning changes	`supporting_files/A_collection/cleaning_changes.csv`	data rows
103 / 1	none_stated / manual prerequisite	`data/processed/courses.json`	`prereq_source`
CS 613	manual prerequisite decision	`data/raw/prereq_manual.csv`	`decided_by`
3	courses with recommended background	`data/processed/courses.json`	non-empty `recommended_background`
104 nodes, 9 edges, 0 cycles, 0 missing	prerequisite graph	`supporting_files/B_engine/prereq_graph_report.json`	`nodes`, `edges`, `edges_by_type`, `cycles`, `missing_prereq_codes`
D4, D5	background decisions	`docs/DECISIONS.md`	rows D4, D5
25 checks	validation suite	`supporting_files/B_validation/processed/results.csv`	25 rows, V01 to V25
raw 10: 7 fail, 2 pass, 1 warn	raw stage	`supporting_files/B_validation/raw/results.csv`	`status`
processed 24 pass, 1 warn (2)	processed stage	`supporting_files/B_validation/processed/results.csv`	`status`; V10 `n_failures`
15 of 15	faults caught	`supporting_files/B_validation/corrupted/faults_caught.json`	`caught`
82 to 80	injected credit fault	same	F05 `description`
1 not injected	semester cap fault	`data/interim/corrupted/faults.json` (output of DVC stage `validate_corrupted`)	`not_injected`
209	corruption seed	`params.yaml`	`seeds.validation_corruption`
Table 1 values	before/after	`supporting_files/B_validation/before_after.md`	table rows
100 of 600	gold changed, MTech-AI fix	`supporting_files/B_engine/gold_fix_mtech_ai_baskets.md`	"Effect"
53; 0	gold changed, query texts changed, D5	`supporting_files/B_engine/gold_fix_ug_background.md`	"Effect"
40 items, 2 rounds	labelling design	`supporting_files/C_labelling/round1/agreement_summary.json`, `round2/agreement_summary.json`	`n_items_compared`
208	annotation order seed	`params.yaml`	`seeds.annotation_order`
1.23.2	Label Studio	`ENVIRONMENT.md`	Python and tools table
Table 2 kappas and CIs	agreement	`supporting_files/C_labelling/kappa_ci.json`	`round*_decision`, `round*_error_category`
2000	bootstrap resamples	same	`n_boot`
65.0 / 62.5; 77.5 / 75.0	raw agreement	`supporting_files/C_labelling/round1/agreement_summary.json`, `round2/...`	`percent_agreement`
17; 12	disagreements	same	`n_disagreements`
34 / 24	A2 / A1 accepts, round 1	`supporting_files/C_labelling/round1/agreement_summary.json`	`label_counts`
11 of 17	A1 flagged, A2 accepted	`supporting_files/C_labelling/round1/disagreement_analysis.md`	"The main pattern"

number	claim	file	field / row
Figure 1	confusion	supporting_files/C_labelling/round1/confusion_decision.csv, round2/confusion_decision.csv	matrix
seven v2 rules	guideline changes	supporting_files/C_labelling/round1/disagreement_analysis.md	R1 to R7
50; 17 + 13	agreed; adjudicated	supporting_files/C_labelling/adjudication_log.json	agreed, adjudicated per round
30 (29 + R2-Q0196)	assistant proposals	supporting_files/C_labelling/adjudication_proposals.json	keys other than _about
28 approve, 1 override	human decisions	supporting_files/C_labelling/adjudication_decisions_export.json	choice, decided_by
1 decided by A1	extra item R2-Q0196	data/labels/audited.jsonl	decided_by "A1 (chose the assistant-drafted combined text in chat)"
0 unresolved	adjudication	supporting_files/C_labelling/adjudication_log.json	unresolved_items
55 / 21 / 4	final accept / edit / reject	data/labels/audited.jsonl	final_decision
27 / 27 / 22	gold by intent	supporting_files/E_splits/gold_summary.json	by_intent
sha256 cd2e08dc...	gold checksum	data/gold/GOLD_SHA256.txt	line 1
Table 3 values; 36 of 120	invariance	supporting_files/D_augmentation/invariance.csv	all rows; n_pass
47 (26 alone) / 35 / 21 / 10	paraphrase failures losing track / semesters / stream / turned to student	same	fail_reasons, counts summed over every reason that names the slot
slot list	semantic check	src/augment/verify.py	slot_issues
336; 232	AL pool; failures	supporting_files/D_active_learning/scores_summary.json	n, failed
94 / 64 / 56 / 18 / 2	failure types	same	failure_types
Table 4 values	AL vs random	supporting_files/D_active_learning/label_efficiency.csv	failures_captured, failure_types_covered, intents_covered
5 seeds	random baseline	params.yaml	seeds.active_learning
stratified added later	AL variant	supporting_files/D_active_learning/summary.md	paragraph after the table
precision 1; not a learning curve	reading of AL	same	"Reading the first column"
26 stages; frozen EE fetch	DVC pipeline	dvc.yaml	stages; fetch_ee_pages: frozen
dvc add inputs	DVC-tracked inputs	data/raw/*.dvc, data/raw/forms/responses_raw.csv.dvc, data/labels/round*/*.dvc	files
../dvc_remote	DVC remote	.dvc/config, README.md	remote "localremote"
a1-final	freeze tag	HANDOFF_A2.md	first paragraph
dvc.lock; byte-for-byte repro	DVC lock and rerun	dvc.lock, data/interim/llm_cache.jsonl.dvc	stage outs hashes
own_programme_baskets	students reproduced exactly	src/engine/rule_engine.py	role_of docstring; supporting_files/B_engine/gold_fix_mtech_ai_baskets.md
207; 0.65 / 0.15 / 0.20	split seed, fractions	supporting_files/E_splits/split_summary.json	seed, fractions
7 of 22	test-only templates	same held_out_templates; queries_summary.json per_template	7 families, 22 templates

number	claim	file	field / row
Table 5 values	split sizes	`supporting_files/E_splits/split_summary.json`	`student_assignment`, `original_rows_per_split`, `child_rows_per_split`, `rows_per_split`, `eligibility_by_split`, `original_rows_by_programme`
all overlaps 0, 0 dropped	split checks	same	`overlap_checks`, `all_overlaps_zero`, `n_dropped_pairs`
22.04.5, 2500U, 3.10.12, 3.67.1	environment	`ENVIRONMENT.md`	Machine, Python and tools tables
T4, CUDA 12.8, 2.11.0+cu128, 5.16.1	Kaggle GPU	`ENVIRONMENT.md`	GPU row
per-prompt model and backends	prompt record	`prompts/pipeline/README.md`	Model, Prompts
prompts 1 to 58	session logs	`prompts/assistant/claude_code_session_2026-10-18.md`	last heading "Prompt 58"
eight seeds	seeds	`params.yaml`	`seeds`
Ollama 0.1.38, llama.cpp, Kaggle T4	LLM backends	`supporting_files/D_augmentation/llm_backend.md`	sections 1, 2, 4
`para_backend` per row	backend record	`data/processed/queries_para.jsonl`	`para_backend`
30	form target	assignment brief (not in the zip)	Section 8, Project 4, Labelling: "at least 30 real queries"
"to be updated soon"	ICDT list incomplete	`supporting_files/A_collection/collection_log.md`, `DATASET_CARD.md`	section 7; limits
34 / 3	commits per author	git history: `git log --format=%an | sort | uniq -c`	hem436 / Myank Singh Beniwal
A2 exports	committed by MSB	git commits f395c8a, 709292f; `LABELLING_PROTOCOL.md`	section 9
form shared by hem436	distribution	`supporting_files/C_labelling/forms_log.md`	row 2
byte-for-byte repro	reproducibility	`supporting_files/F_reproducibility/repro_check.txt`	`sha256sum -c` lines; `dvc status`
252	tests passing	`supporting_files/F_reproducibility/pytest.txt`	last line
12 of 180	students the AI fix would change	`supporting_files/B_engine/gold_fix_mtech_ai_baskets.md`	last section

B AI assistance

What the assistant did. Claude Code (model claude-opus-5-5) wrote most of the code through subagents. Their prompts are in `prompts/assistant/`: `agent_common_rules.md`, `agent_A_curriculum.md`, `agent_A2_ee_pages.md`, `agent_B_engine.md`, `agent_B2_engine_revision.md`, `agent_C_labelling.md`, `agent_D_validation.md`, `agent_E_llm_augment_al.md`, `agent_F_splits_dvc_docs.md`, `agent_F2_resplit.md` and `agent_G_report.md` (this report) and `agent_H_mock_grading.md` (mock grading). The user's own prompts are logged in `claude_code_session_2026-10-17.md` and `claude_code_session_2026-10-18.md`, which cover prompts 1 to 58. The assistant also drafted the 30 adjudication proposals (a human made every final call), built the labelling kit pages and the adjudication review page, and built and edited the Google Form in the browser (Claude in Chrome) at the user's request.

Pipeline LLM. Qwen2.5-1.5B-Instruct (`qwen2.5:1.5b`, Q4_K_M via llama.cpp, or fp16 via trans-

formers on Kaggle) paraphrases queries and justifications and gives base-model answers. Its prompts are files in `prompts/pipeline/`: `paraphrase_query_v1.txt`, `paraphrase_justification_v1.txt`, `augment_paraphrase_v1.txt`, `base_model_answer_v1.txt`. Model, digest and backend are stored with every output, and `prompts/pipeline/README.md` lists model, versions and backends per prompt and the Qwen default system prompt.

What humans did. All labelling (A1 and A2, both rounds), every adjudication decision, the scope and modelling decisions in `docs/DECISIONS.md`, both gold-bug findings, and form sharing and distribution. The assistant declined to make commits under Myank's identity; his commits are his own.

C Guidelines v2 and label levels

Label levels, in this order: 1) decision: accept, edit, reject; 2) error category (one, by priority: hallucinated_course, wrong_fact_or_policy, meaning_drift, incomplete_justification, ambiguous_query, fluency; or none); 3) edit text, only for edit.

Changes from v1 to v2 (`labelstudio/guidelines_v2.md`, sources in round-1 `disagreement_analysis.md`):

- R1: a justification must state the gold outcome and every number, course and open item; pure encouragement is incomplete_justification.
- R2: any number or status that contradicts gold or is not in gold is wrong_fact_or_policy, including vague text that implies a wrong status.
- R3: edit when the fix touches at most two sentences or one query phrase, else reject; a missing fact alone is never a reason to reject.
- R4: added scope, "available" for "remaining", or a question turned back on the student is meaning_drift.
- R5: with two problems, pick the category by priority and fix both texts.
- R6: gold is the truth; do not correct it from memory, flag "GOLD?" instead.
- R7: formatting and tone are not errors; edit or reject always needs a category other than none.

The label set and the Label Studio config (`labelstudio/config_v1.xml`) did not change.